1. (a) Sketch the circuit of a VGS multiplier with equations describing its operation.

[4]

V_{GS} Multiplier

Stolen from the V_{BE} multiplier

Circuit Analysis

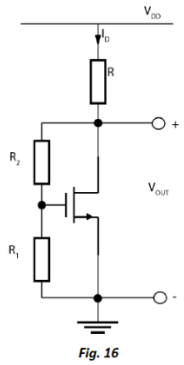


Fig. 16

$$V_{OUT} = V_{GS}\left[1+\frac{R_2}{R_1}\right]$$

OR

$$V_{OUT} = \left[1+\frac{R_2}{R_1}\right]\left(V_T + \sqrt{\frac{V_{DD}}{\beta R}}\right)$$

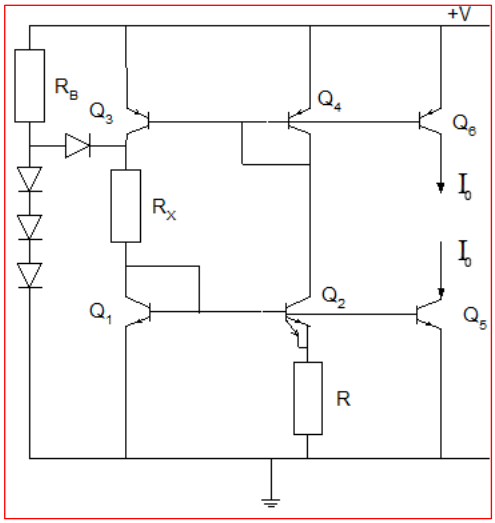
Note: All R's can be replaced with active resistors

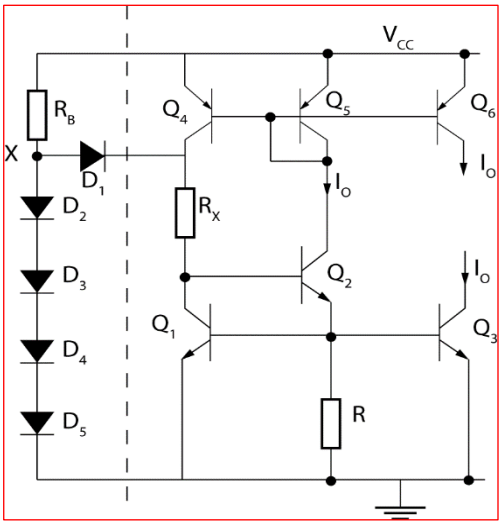
25

[4]

(b) Sketch the circuits of two supply independent current sources, including circuitry to guarantee their operation on start-up and explain their main differences.

[6]





[6]

(c) Using the equation for fractional temperature coefficient, explain which of the two circuits in 1. (b) is more insensitive to temperature.

[5]

Must elaborate that one circuit has a better temperature coefficient that that the other and show equations for Temperature coefficient.

Elimination of Vbe produces an Improved TCF using VT Referenced Current Source/Sink

$$TC_F = \left(\frac{1}{V_T}\frac{dV_T}{dT}\right) - \left(\frac{1}{R}\frac{dR}{dT}\right)$$

A V_{be} Referenced Sink/Source with a temperature coefficient has worse performance:

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$$TC_F = \left(\frac{1}{V_{be1}}\frac{dV_{be1}}{dT}\right) - \left(\frac{1}{R}\frac{dR}{dT}\right)$$

[5]

(d) Sketch a suitable two transistor and four transistor potential divider voltage reference circuit and show that when VDD=+5V, VSS=-5V and ID=10uA, for an output voltage Vout=0, the four transistor circuit consumes less area than the two transistor one. You may assume K_P=8μA/V², K_N=17μA/V², V_{TN}=V_{TP}=1V.

[5]

The two and four transistor potential dividers depicted in figure below.

$$K_n = 17\mu A/V^2; K_p = 8\mu A/V^2; V_{Tn} = V_{Tp} = 1V; V_{DD} = +5V; V_{SS} = -5V; V_{Out} = 0$$

For the two-transistor potential divider

$$I_{Qp} = I_{Qn} = \tfrac{1}{2} K_n \frac{W_{Qn}}{L_{Qn}} (V_{out} - V_{ss} - V_{Tn})^2 = \tfrac{1}{2} K_p \frac{W_{Qp}}{L_{Qp}} (V_{out} - V_{dd} - V_{Tp})^2$$

$$\Rightarrow \frac{W_{Qn}}{L_{Qn}} = \frac{10}{136} = 0.073; \frac{W_{Qp}}{L_{Qp}} = \frac{10}{64} = 0.156$$

For the four-transistor potential divider

$$I_{Qp1} = I_{Qn1} = I_{Qp2} = I_{Qn2} = \tfrac{1}{2} K_n \frac{W_{Qn1}}{L_{Qn1}} (V_{GSn1} - V_{Tn})^2 = \tfrac{1}{2} K_n \frac{W_{Qn2}}{L_{Qn2}} (V_{GSn2} - V_{Tn})^2$$

$$= \tfrac{1}{2} K_p \frac{W_{Qp1}}{L_{Qp1}} (V_{GSp1} - V_{Tp})^2 = \tfrac{1}{2} K_p \frac{W_{Qp2}}{L_{Qp2}} (V_{GSp2} - V_{Tp})^2$$

$$V_{GSn1} = V_{GSn2} = \frac{V_{Out} - V_{SS}}{2} = 2.5V$$

$$V_{GSp1} = V_{GSp2} = \frac{V_{Out} - V_{DD}}{2} = 2.5V$$

$$\frac{W_{n1}}{L_{n1}} = \frac{W_{n2}}{L_{n2}} = \frac{10}{19.125} = 0.523$$

$$\frac{W_{p1}}{L_{p1}} = \frac{W_{p2}}{L_{p2}} = \frac{10}{10} = 1$$

The aspect ratio of the transistors in the four-transistor potential divider is much less than the two-transistor circuit, therefore, its consuming area will be less. For example, assuming common width of 10 units, for the two-transistor configuration the active area becomes 10*(64+136) =2000 units while for the four transistor it becomes 10*2*(10+19.125) =582.5 units.

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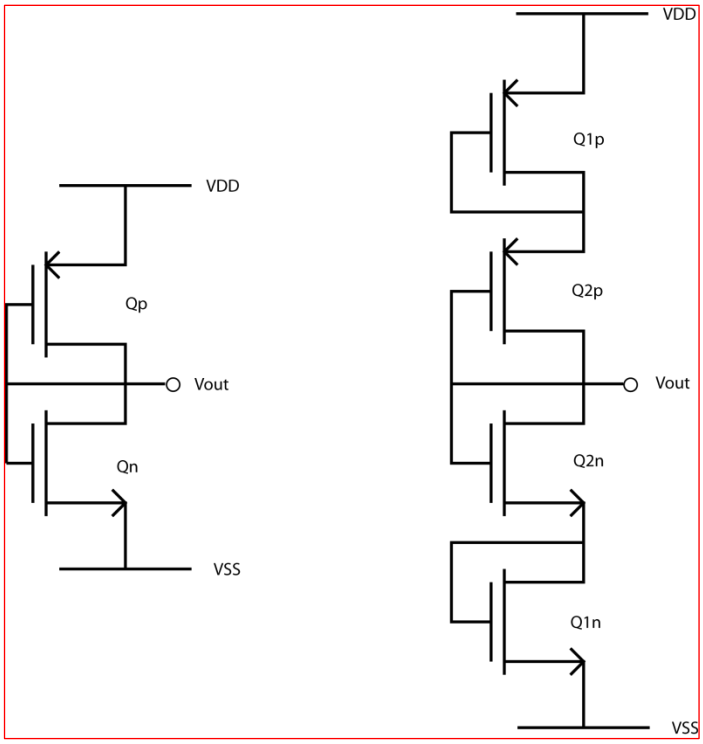


Figure. Answer to Question 1. (d); Potential divider circuits

[5]

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2. Figure 2 shows a partially-designed two-stage CMOS op-amp required to give a voltage gain of approximately 80 dB, a slew-rate of 3 *V/μs* and a gain-bandwidth product of 3 MHz.

(a) Given that the technology is a fixed 5 μm double metal CMOS process, design the channel widths of transistors Q1, Q2 and Q6 for the op-amp to meet the above performance specifications. Aspect ratios of all other devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below.

[12]

$$I_o = I_{d6} = 20e^{-6}$$
$$I_{d1} = I_{d2} = \frac{I_o}{2} = 10e^{-6}$$

$$Slew\ Rate = \frac{I_o}{C_c} \rightarrow C_C = \frac{I_o}{Slew\ Rate} = \mathbf{6.67pF}$$
$$GBP = \frac{g_{m2}}{2\pi C_c} \rightarrow g_{m2} = GBP \times 2\pi C_c = \mathbf{1.257\ e^{-4}\ A/V}$$
$$g_{m1} = g_{m2} = \sqrt{\beta_2 I_{d2}} \rightarrow \beta_2 = \beta_1 = \mathbf{15.8e^{-4}} = \frac{K_n W}{2L}$$

$$\left(\frac{\boldsymbol{W}}{\boldsymbol{L}}\right)_{\mathbf{1}} = \left(\frac{\boldsymbol{W}}{\boldsymbol{L}}\right)_{\mathbf{2}} = \mathbf{105} = \frac{\mathbf{525\mu}}{\mathbf{5\mu}}$$

$$(g_{o2} + g_{o4}) = I_{d2}(\lambda_n + \lambda_p) = \mathbf{5e^{-7}\ \Omega^{-1}}$$
$$A_1 = \frac{g_{m2}}{(g_{o2} + g_{o4})} \rightarrow A_1 = \mathbf{251}$$
$$A_2 = \frac{1e^4}{A_1} \rightarrow A_2 = \mathbf{39}$$

$$(g_{o6} + g_{o7}) = I_{d6}(\lambda_6 + \lambda_7) = \mathbf{1e^{-6}\ \Omega^{-1}}$$

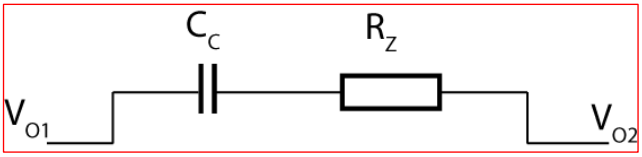
$$A_2 = \frac{g_{m6}}{(g_{o6} + g_{o7})} \rightarrow g_{m6} = \mathbf{3.9e^{-5}\ A/V}$$
$$\beta_6 = \left(\frac{g_{m6}}{2}\right)^2 \frac{1}{I_{d6}} = \mathbf{1.9e^{-5}} = \frac{K_p W}{2L}$$
$$\left(\frac{\boldsymbol{W}}{\boldsymbol{L}}\right)_{\mathbf{6}} = \mathbf{1.9} = \frac{\mathbf{19\mu}}{\mathbf{10\mu}}$$

[12]

(b) Show with equations and describe why a resistor in series with the compensation capacitor *C* in Figure 2 can significantly improve the amplifier's phase margin.

[4]

Introducing a nulling resistor boot straps the capacitor and significantly increases the phase margin by shifting the second pole. This can also be selected such that the zero introduced cancels out the second pole.



Effectively using a resistor to try and bootstrap C_C, so that at high frequencies it breaks the feed forward path. This is achieved by making

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$R_Z = 1/g_{m6}$ so that the voltage across the resistor will cancel the signal voltage on the left hand plate of the capacitor.

The location of the RHP zero however, changes to

$$Z = \frac{1}{C_C\left[\frac{1}{G_{m2}} - R_Z\right]} = \frac{1}{C_C\left[\frac{1}{g_{m6}} - R_Z\right]}$$

$$f_Z = \frac{1}{2\text{p}C_C\left[\frac{1}{g_{m6}} - R_Z\right]}$$

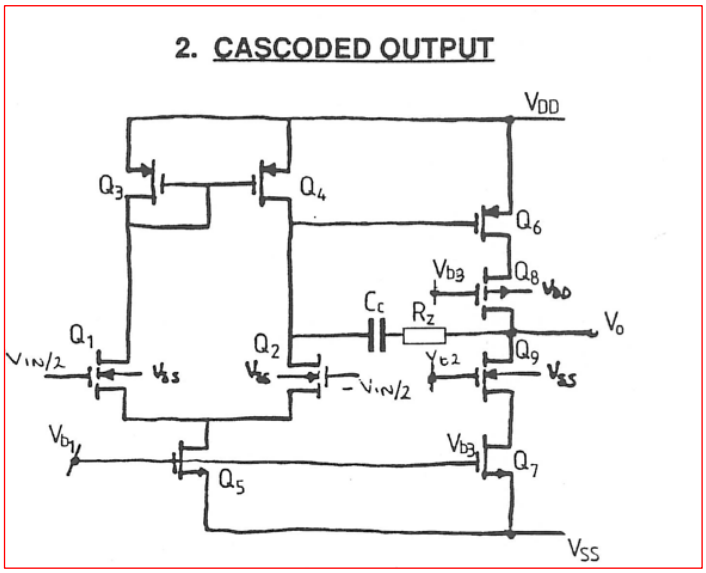
Or

We can cancel Z and push f_Z to infinity by simply making $\boxed{R_Z = \frac{1}{g_{m6}}}$

[4]

(c) Discuss any limitation to the PSRR performance of this circuit and sketch an updated circuit that obtains better PSRR performance.

[4]



Limited impedance seen from the output to power supply will result in a limited PSRR performance.

Solution: Cascoded output circuit for improved PSRR

[4]

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3. (a) Give one advantage and one disadvantage of each of the two CMOS current mirror circuits shown in Figures 3.1 and 3.2.

[4]

a) Figure 3.1:
Advantage: Large voltage headroom/good for low voltage operation as has needs only one ΔV on output to be in saturation.
Disadvantage: Errors on output current due to ΔVDS will be significant.

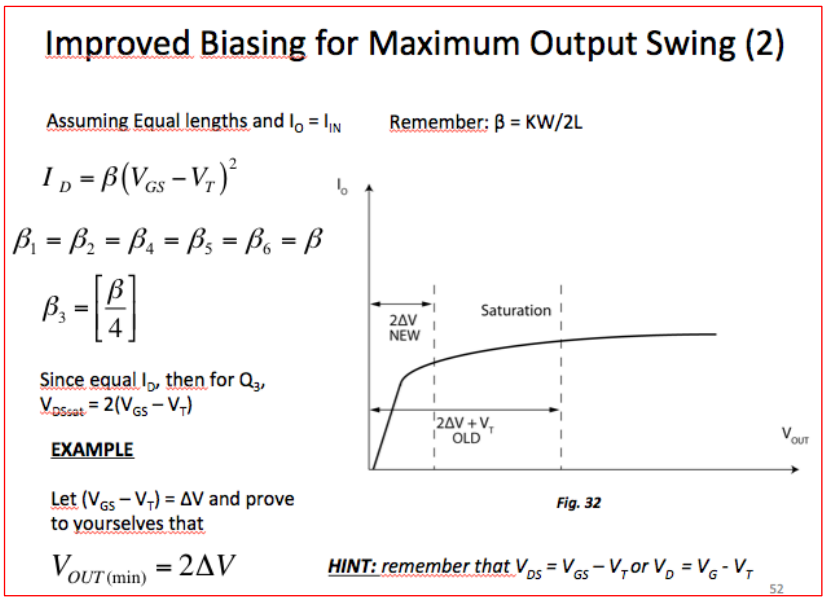
Figure 3.2:
Advantage: Improved output impedance due to cascoding and therefore less error due to early effect, changes in ΔVDS
Disadvantage: Two output transistors reduce output voltage headroom because circuit needs two ΔV to keep output transistors in saturation.

[1 point for each, total 4 points]

(b) For the current mirror of Figure 3.2 derive this voltage swing in terms of device threshold voltage V_T, clearly stating any assumptions you make.

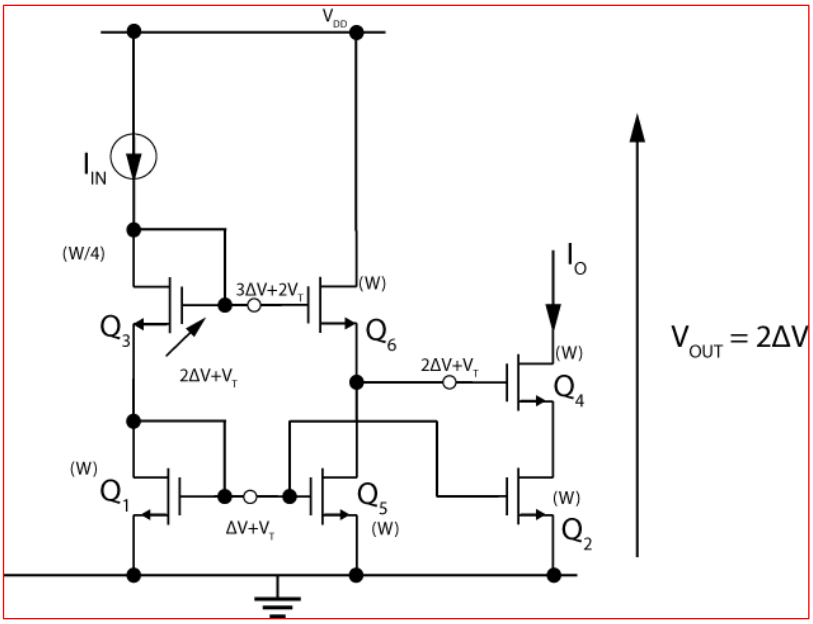
[6]

b) From notes:



Annotation of VGS' of each transistor:

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[6]

(c) (i) Show how a 2-transistor current mirror can be used as the active load of a long-tail pair and derive the differential voltage gain.

[4]

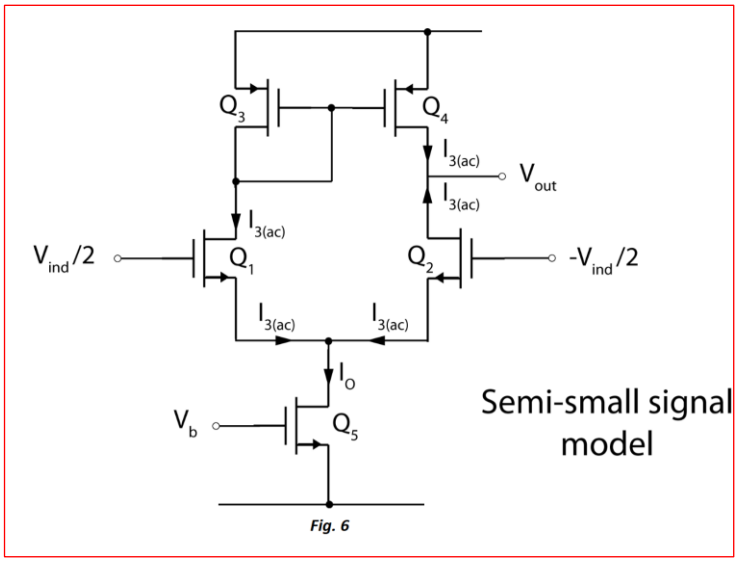


Fig. 6

The output voltage is:

$$V_d = 2\left(\frac{g_{m12}V_{ind}}{2}\right)(r_{o2} \parallel r_{o4})$$

The differential voltage gain is

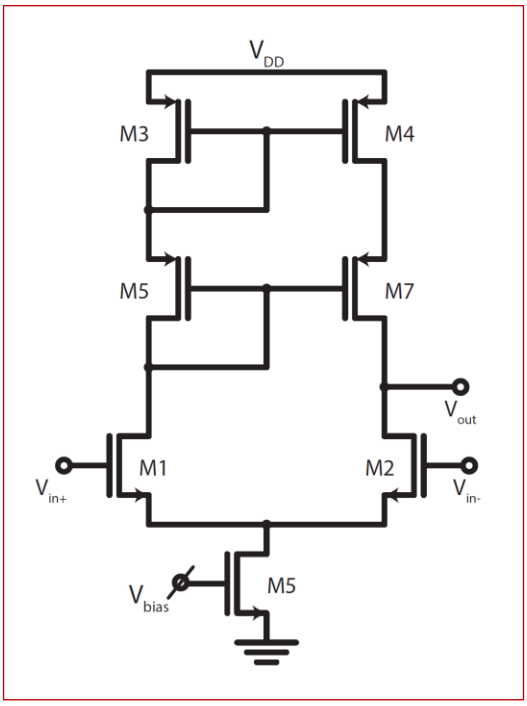
$$A_d = g_{m12}(r_{o2} \parallel r_{o4}) \text{ or } A_d = \frac{g_{m12}}{g_{o2}+g_{o4}}$$

[4]

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(ii) Replace the current mirror to a cascode variant and show the revised voltage gain equation.

[4]



The output impedance is:

$$R_o = r_{o2} \parallel [r_{o7}(1 + g_{m7}r_{o4})] \approx r_{o2} \parallel g_{m7}r_{o7}r_{o4}$$

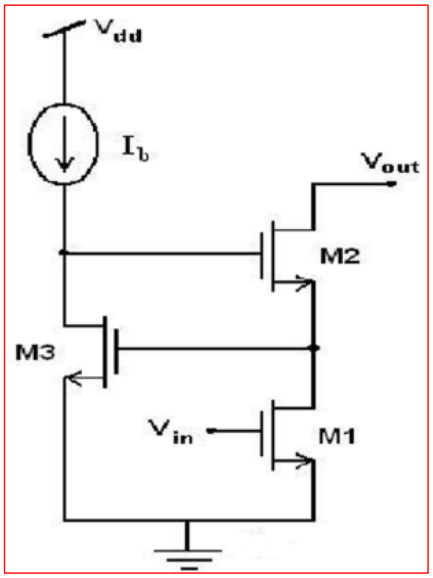
The differential voltage gain is

$$A_d = g_{m12}(r_{o2} \parallel g_{m7}r_{o7}r_{o4}) \text{ or } A_d = \frac{g_{m12}}{g_{02}+g_{04}(\frac{g_{07}}{g_{m7}})}$$

[4]

(d) Sketch a regulated cascode current mirror.

[2]



[2]

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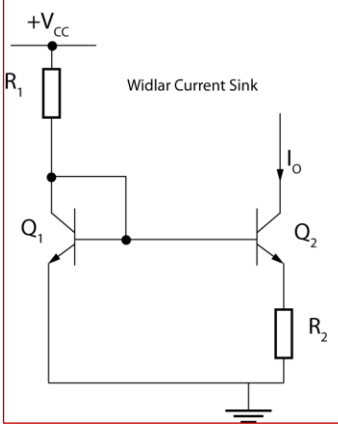
4. (a) Figure 4.1 shows a commonly used circuit for providing a stable current reference. The circuit has the following values I_O = 5μA, R_1 = 14.4KΩ, I_{IN} = 1mA, R_2 = 26.5KΩ. R_2 is a diffused resistor with a temperature coefficient of +1500ppm/°C at T=300K. The output current sensitivity to the supply voltage V_{cc} is given by:

$$S_{V_{CC}}^{Io} = \frac{1}{1+\frac{I_O R_2}{V_T}} S_{V_{CC}}^{I_{IN}}$$

(i). Derive the term S_{VCC}^{Iin} and thus the output current sensitivity.

[5]

EXAMPLE



$$I_O = \frac{\Delta V_{be}}{R_2} \quad \text{So} \quad I_O = V_T \frac{\ln\left[\frac{I_O}{I_{IN}}\right]}{R_2}$$

Can Show:

$$S_{V_{CC}}^{Io} = \frac{1}{1+\frac{I_O R_2}{V_T}} S_{V_{CC}}^{I_{IN}}$$

$$I_{IN} = \frac{(V_{CC} - V_{be})}{R_1} \therefore S_{V_{CC}}^{I_{IN}} = \frac{V_{CC}}{V_{CC} - V_{be}}$$

Since $V_{CC} >> V_{be}$ then $\boxed{S_{V_{CC}}^{I_{IN}} = 1}$

[5]

(ii). Calculate the change in I_o if VCC where to change by 20%.

[5]

A 1% change in V_{CC} causes a 1% change in I_{IN}

$$S_{V_{CC}}^{Io} = \frac{1}{1+\frac{I_O R_2}{V_T}}$$

EXAMPLE

I_O = 5μA, R_1 = 14.4KΩ, I_{IN} = 1mA, R_2 = 26.5KΩ

Thus $$S_{V_{CC}}^{I\,o} = \frac{1}{1+\frac{5\times10^{-6}\times26.5\times10^{3}}{25\times10^{-3}}}$$

$$S_{V_{CC}}^{I_o} = 0.16$$

Therefore a 20% change in V_{CC} will cause I0 to change by 3.2%.

[5]

(b). Figure 4.2 shows a differential opamp which is based upon a single-stage fully differential design.

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(i). Transistors Q1 and Q2 of Figure 4.2 could be considered to form a linear differential resistor. Derive the equation for linear differential resistance. How does the body effect affect the linearity of each of these transistors?

[5]

$$I_1 = 2b[(V_g - V_x - V_T)(V_{in}/2 - V_x) - 1/2(V_{in}/2 - V_x)^2]$$
$$I_2 = 2b[(V_g - V_x - V_T)(-V_{in}/2 - V_x) - 1/2(-V_{in}/2 - V_x)^2]$$
$$R = 2V_{in}/(I_1 - I_2)$$
$$R = \frac{1}{b(V_g - V_T)}$$

[5]

(ii) Assuming the time constant of the circuit is τ=10ms, calculate the value of feedback capacitor, C, assuming input transistors Q1 and Q2 are operating in the triode region and have W/L=5, K=20uA/V2, Vg=1V, Vt=0.5V and node x is at a common mode of 0 V. You may neglect any body effects.

[5]

Time constant τ=RC=10ms
R=1/20ux5(1-0.5)/2=40 KΩ
C=τ/R=10m/40K=2.5x10^{-7} f

[5]

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5. (a) Figure 5.1 shows one bit of an algorithmic data converter.

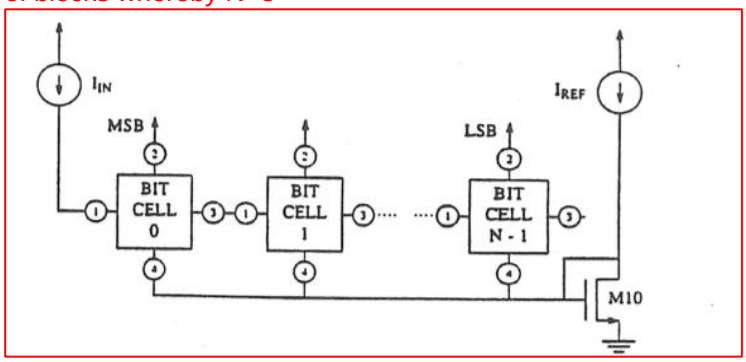
(i) Explain how the data conversion works.

[3]

(ii)Show how this can be cascaded to perform an 8-bit data conversion.

[4]

Draw a cascade of blocks whereby N=8



[4]

(b) What noise limits the resolution of sampled-data converters?

[3]

$$DR = 2^N = V_{ref}/Noise = V_{ref}/\sqrt{(kT.10Rf_c)}$$

[3]

(c) Show how the integrator of Figure 5.2 can be used to mimic both the inductor and capacitor of an LCR passive ladder filter.

[10]

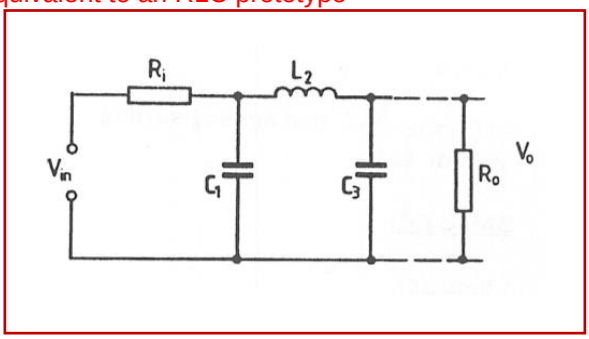
General transformation rules for ladder prototypes:

Inductor: $\frac{f_c L_2}{R_s} = \frac{C_{L2}}{C_u}$

Capacitor: $\frac{C_{c3}}{C_u} = f_c R_s C_3$

Resistor Rs=dummy scalar.

The circuit is equivalent to an RLC prototype

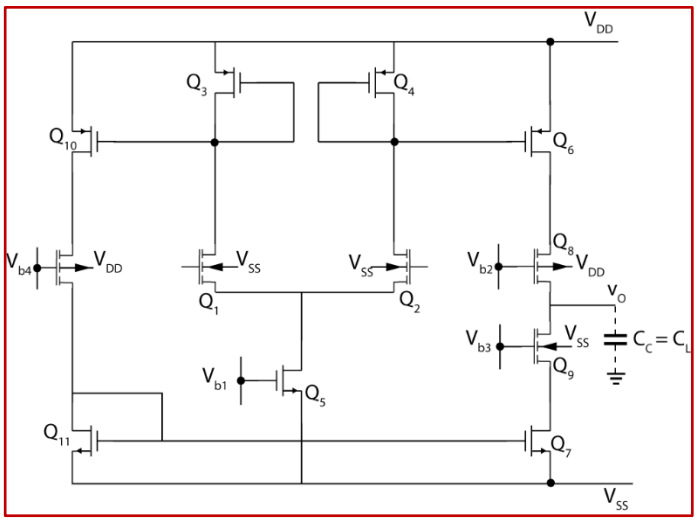


[10]

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6. (a) Sketch the circuit of single-stage differential to single ended output high frequency operational amplifier that does not require miller compensation.

[5]

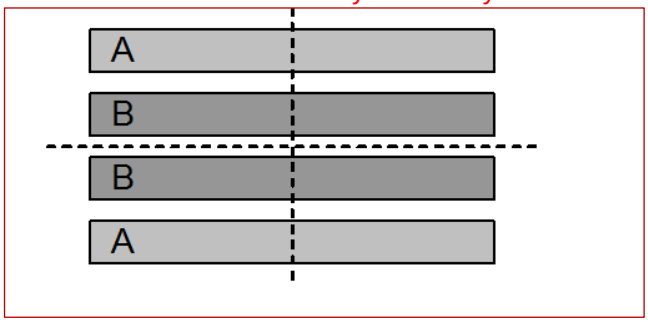


[5]

(b) When matching transistors special care is taken with the layout. Discuss what is meant by common centroiding when matching two transistors.

[5]

Mismatches like λ (the Vds difference), threshold voltage VT and W/L may cause errors in offset and gain. A proper layout may help reduce the effect of such mismatches as well as gradients. Common centroid design splits the transistors over a common centroid or axis of symmetry. For example, the elements A and B in an array may be inter-digitated as in figure below. The input differential pair transistors in the Op-amp are the critical ones for the layout as they affect the offset and gain.



[5]

(c) For fully differential output op-amps explain what is meant by common mode feedback using a diagram where necessary.

[5]

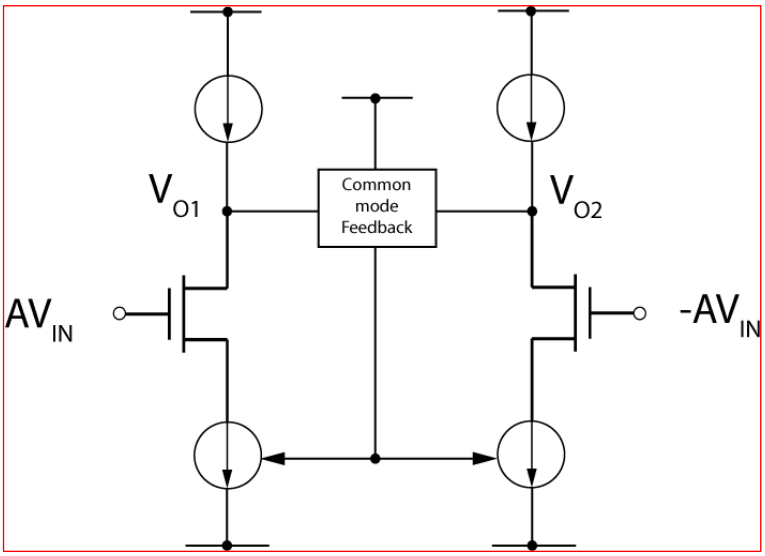
If an amplifier has a differential output with high gain, the voltages at the outputs are pretty well ill defined and will drift all over the place, usually taking the amplifier out of its high gain region.

We therefore need to sample an output which is “common” to both and stabilise this voltage.

The technique is known as common-mode feedback and we will describe it in slightly more detail when we come on to differential in, differential out, CMOS op-amp architectures.

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We will show here how the technique can be used to improve the CMRR of the simple differential amplifier.

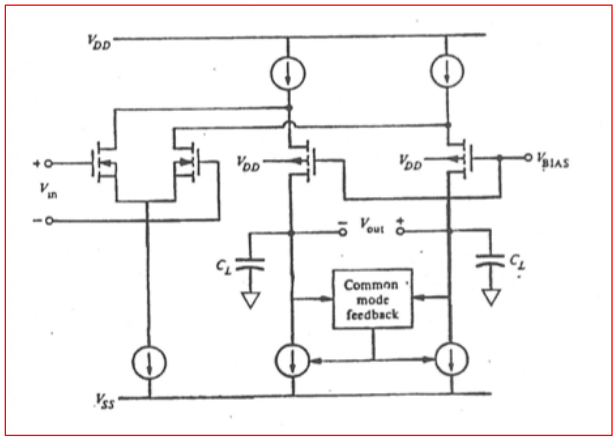


[5]

(d) Sketch the circuit of a differential operational amplifier that has a single dominant pole and uses common mode feedback to stabilise the output DC voltage.

[5]

In the folded cascode we are unstacking the conventional cascode and spreading it out. The AC current path is folded, and this allow a reduction in power supply. The conventional cascode requires a 2-stage architecture and since the impedance at (x) is high requires internal compensation. The folded cascode can be used as a single stage architecture, node x is low impedance so the only high impedance node will be at the output.
Gain of the folded cascode Av=gm1/Go.



[5]

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